

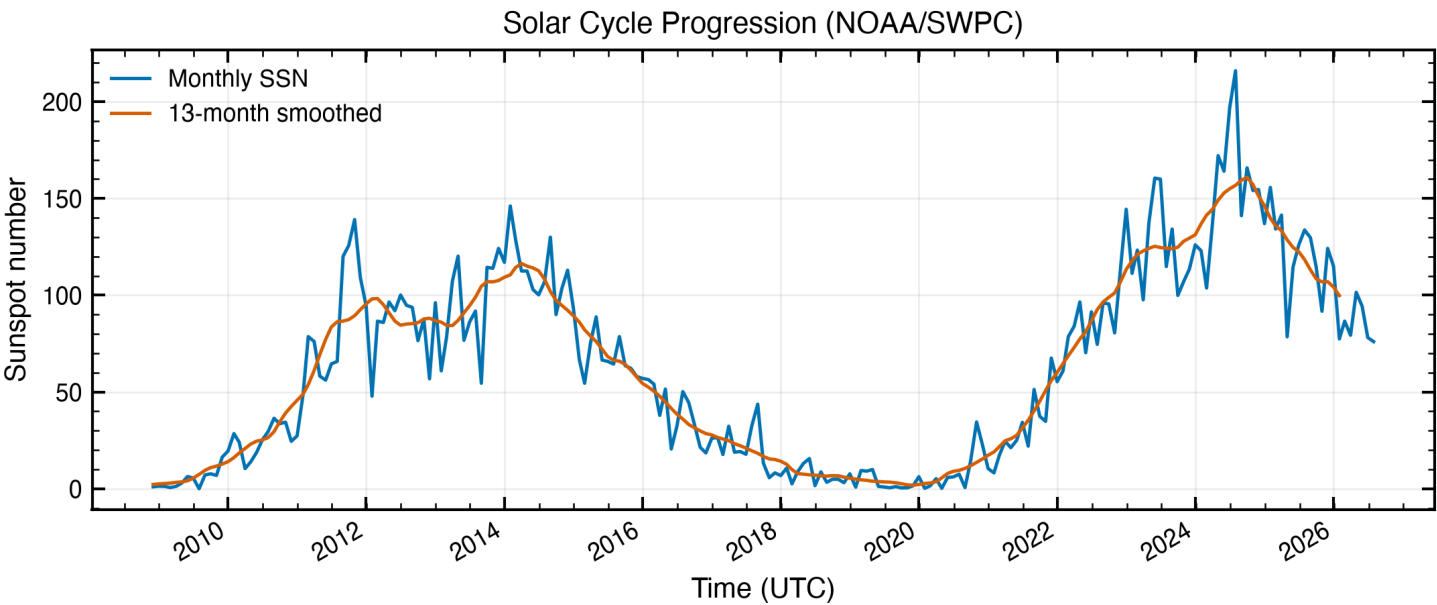
Reporting window and provenance

Window: 2026-09-02T11:00:00 to 2026-09-03T11:00:00 UTC. Sources: GOES primary XRS 1-min, NOAA RTSW solar wind at L1, NOAA planetary Kp, NOAA/SWPC solar region analysis, NASA CCMC DONKI, NOAA Solar Cycle Progression. Operational feeds - not science quality; every number is audit-logged (workspace/logs/audit.jsonl):

fetch_swpc_timeseries: ok (d8a8a5bc3670)
find_flares: ok (38738edf705f)
fetch_swpc_timeseries: ok (898509b36723)
fetch_swpc_timeseries: ok (f58ebd8ba6df)
fetch_swpc_timeseries: ok (fac00a1b9cc4)
get_solar_regions: ok (254a1a5166fc)
search_donki: ok (fcde05139e00)
fetch_solar_cycle: ok (645562724fef)
describe_series: ok (208869606520)
describe_series: ok (3dfd922de967)
plot_timeseries: ok (55b95d67074f)
plot_stack: ok (8001467645e8)
plot_timeseries: ok (020b34caa9d2)

Solar cycle

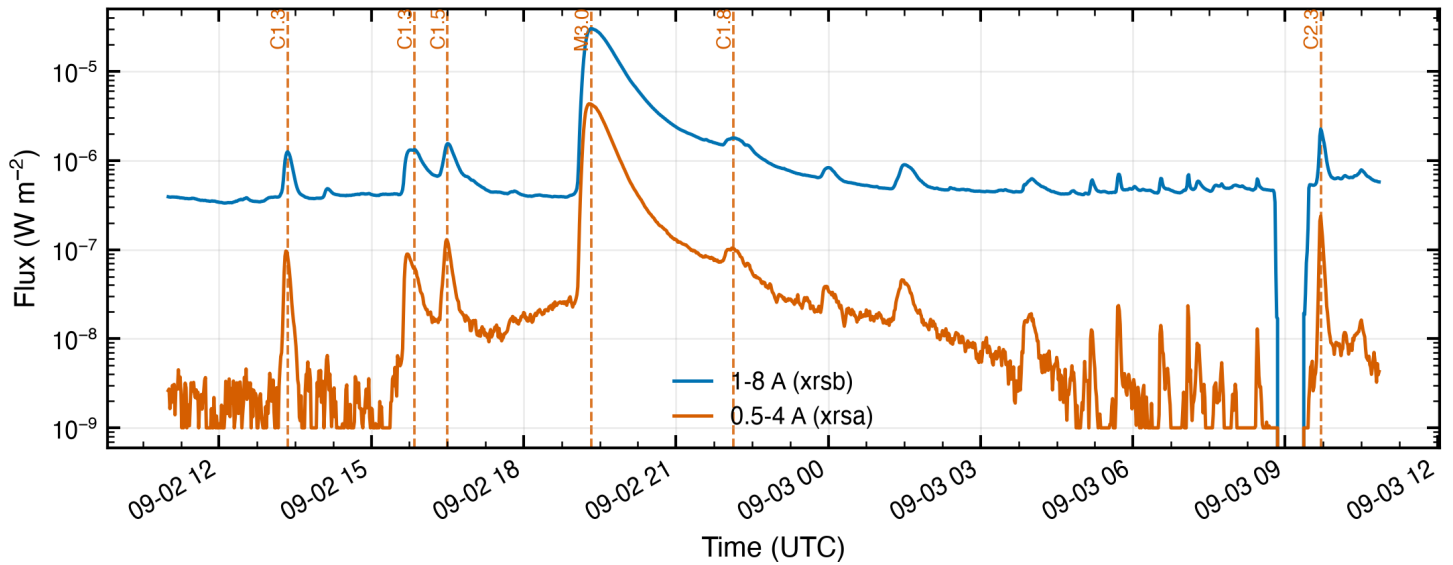
Latest monthly sunspot number: 76.0 (2026-08; previous month 78.1). Cycle 25 smoothed maximum 160.9 in 2024-10 (smoothed series final through 2026-02).



Flare activity

Detected 0 X, 1 M, 5 C flares (plus 33 B-level enhancements; sensitive detector - the official SWPC event list applies stricter criteria). Strongest: M3.0 (peak 19:20 UT, 57 min).

GOES XRS (operational), 2026-09-02 11:00 to 2026-09-03 11:00 UT



Sunspot regions (NOAA analysis)

Analysis date 2026-09-03:

AR4523 N11E35 mag A 3 spots, area 40 mu
 AR4520 S12W47 mag A 1 spots, area 10 mu
 AR4524 N12E64 mag A 1 spots, area 60 mu
 AR4521 N09E25 mag A 1 spots, area 80 mu

CMEs (DONKI)

2026-09-02T00:08:00-CME-001: Narrow CME first seen to the W in STEREO A COR2 beginning at 2026-09-02T00:08Z. No appropriately timed source can be seen on the solar disk. There is

2026-09-02T07:24:00-CME-001: CME first seen to the E in SOHO LASCO C2 starting around 2026-09-02T07:24Z. It is also seen to the E in SOHO LASCO C3 and GOES CCOR-1 and to the SE in

2026-09-02T10:24:00-CME-001: Very faint CME first seen to the E in SOHO LASCO C2 starting around 2026-09-02T10:24Z. It is more easily seen to the E in STEREO A COR2. It is not see

2026-09-02T19:36:00-CME-001: Bright CME visible to the NE in SOHO LASCO C2, C3 and GOES CCOR-1 imagery. This CME is not visible in STEREO A COR2 imagery due to a data gap. The sou

Solar wind and geomagnetic conditions

Proton speed 330-449 km/s (mean 376); density mean 5.3 /cm³. Bt 1.4-8.7 nT; Bz GSM -4.5 to 6.8 nT. Kp 0.67-2.33.

Solar wind at L1 (RTSW) + Kp, 2026-09-02/2026-09-03

